

VANSHITA ARYA

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Core Competencies

Data Science & Machine Learning

Cybersecurity & Fraud Analytics

Big Data & Cloud Computing

Data Engineering & Processing

Statistical Analysis & Forecasting

NLP & Text Analytics

Data Visualization & Reporting

Model Deployment & Automation

Financial & Risk Analytics

Regulatory Compliance & Governance

Technical Skills

Programming Languages: Python | SQL | R

Machine Learning & AI: Scikit-learn | TensorFlow | PyTorch | NLP | ML

Data Visualization: Power BI | Matplotlib | Seaborn

Database Management: SQL | PostgreSQL | NoSQL

Big Data & Cloud: Dask | Hadoop | Spark | Azure Fundamentals

Tools & Frameworks: Git | Excel | Flask | Streamlit | CBS Finnacle

Education

- 2023 - Present
M.Sc. in Data Science
Amity University, Noida
CGPA: 9.13 (till 3rd SEM)
- 2017 - 2020
B.Sc. in Maths, Statistics, Physics
Banasthali Vidyapith
CGPA: 7.6

Social Links

[GitHub](#)
[kaggle](#)
[Hackerrank](#)
[Newsletter](#)

Profile Summary

- Data Science Professional with Hands-On Experience:** Strong background in machine learning, predictive analytics, and fraud detection, acquired through industry internships and academic projects.
- Expert in Cyber Fraud Analysis & Financial Crime Investigation:** Assisted law enforcement and financial institutions in identifying fraud patterns and money laundering risks using advanced data techniques.
- Proven Ability to Develop Impactful Data Solutions:** Designed and implemented data-driven systems such as fraud response automation and stock market analytics.
- Passionate About Data-Driven Decision Making:** Skilled in deriving meaningful insights from complex datasets, enabling businesses and agencies to make informed choices.
- Collaborative & Research-Oriented Mindset:** Adept at working in interdisciplinary teams and continuously learning new methodologies in AI and analytics.

Work Experience

Dec'24 - Present | **Cyber Crime Data Analyst Intern** | **I4C, Ministry of Home Affairs**

- Cyber Fraud Analysis & Investigation:** Analyzed **NCRP portal data** to detect fraud patterns in **UPI, e-wallets, cryptocurrency, and CBDC**. Identified **money laundering hotspots** and assisted in **cross-verifying cyber complaints** for risk mitigation.
- Data Extraction & Visualization:** Aggregated **financial data** from major banks and built **Power BI dashboards** to track **ATM withdrawal hotspots, fraud trends, and bank performance**.
- Geospatial & NLP Processing:** Used **geospatial mapping** to analyze fraud locations and applied **NLP & RegEx** to extract regional insights from complaint data.
- Collaboration & Compliance:** Partnered with **Telecom Service Providers (TSPs)** to **block fraudulent mobile numbers** and worked with **banks & law enforcement agencies** to strengthen cybersecurity frameworks.
- Cyber Commando Program Support:** Assisted **police officials** in onboarding the **Cyber Commando Program**, resolving **technical issues**, and maintaining **participant records**.
- Additional Contributions:** Designed an **Anti-Scam Quiz** to enhance fraud awareness. Drafted **DO Letters & Reports** summarizing **fraud trends and bank performance**.

Mar'24 | **Data Science Intern** | **British Airways**

- Extracted & Preprocessed** 10K+ customer reviews using **BeautifulSoup (web scraping)**, **tokenization**, **lemmatization**, and **stopword removal**.
- Applied Sentiment Analysis** using **VADER & TextBlob** to detect customer sentiment trends.
- Used Word2Vec & Topic Modeling** to analyze customer feedback and generate insights.
- Developed a Random Forest Model** to predict customer holiday purchases, improving classification accuracy.
- Built Data Visualizations** using **Seaborn & Matplotlib** to present findings for better business decisions.

Projects

Fraud Intelligence & Response Automation (FIRA) | Python, Excel, Power BI:

- Developed an automated **fraud detection system**, improving detection efficiency by **40%**.
- Created a real-time **complaint tracking dashboard** for financial fraud cases, enhancing response time.

National Stock Exchange Data Analysis | Python, Streamlit, Kaggle API

- Designed a **self-service BI platform** for analyzing stock performances in NSE India.
- Published a **Kaggle dataset** with weekly updates, achieving a high usability score of **8.82**.

Certificate and Courses

- Azure Fundamentals** (Microsoft)
- Introduction to Git and GitHub** (Google)
- Introducing Multimodal Llama 3.2** (Meta)
- Essentials of Data Science with R – Probability and Statistical Inference** (IIT Kanpur)
- Everyday Excel** (University of Colorado Boulder)